HW1

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Annualized Geometric Mean Returns

- US equity dominates long-run performance
- HY credit sits between bonds and equity (as expected)
- Commodities underperform despite high volatility
- EU assets generally underperform US counterparts

	Start	Ann. Geo. Mean Return (%)
US 10Y Bond	2000-02	3.60
EU 10Y Bond	2000-02	2.96
US Equity	2000-02	8.39
EU Equity	2000-08	4.75
US HY Credit	2007-05	4.92
EU HY Credit	2008-02	0.46
Commodities	2006-03	2.14

Table: Annualized Mean Return

Risk vs Return Tradeoff

EU Bond: best, highest return/vol ratio

Commodities: worst risk-adjusted asset

Bonds outperform on risk-adjusted basis due to falling rates

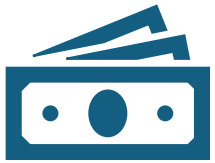
US assets outperform EU in most categories

HY credit risk understated by volatility (tail risk)

	Ann. Geo. Mean (%)	Ann. Volatility (%)	Return / Vol
US 10Y Bond	3.60	7.60	0.474
EU 10Y Bond	2.96	5.09	0.581
US Equity	8.39	15.19	0.552
EU Equity	4.75	21.06	0.225
US HY Credit	4.92	10.28	0.478
EU HY Credit	0.46	3.73	0.124
Commodities	2.14	18.54	0.115

Table: Risk and Return Summary

Decomposition of Returns and Risk



Bonds = carry-driven returns, price-driven risk

Equity = price-driven returns AND risk

	Total Return (%)	Carry / Income (%)	Price / Capital (%)
US 10Y Bond	3.60	3.36	0.24
US Equity	8.39	1.87	6.41

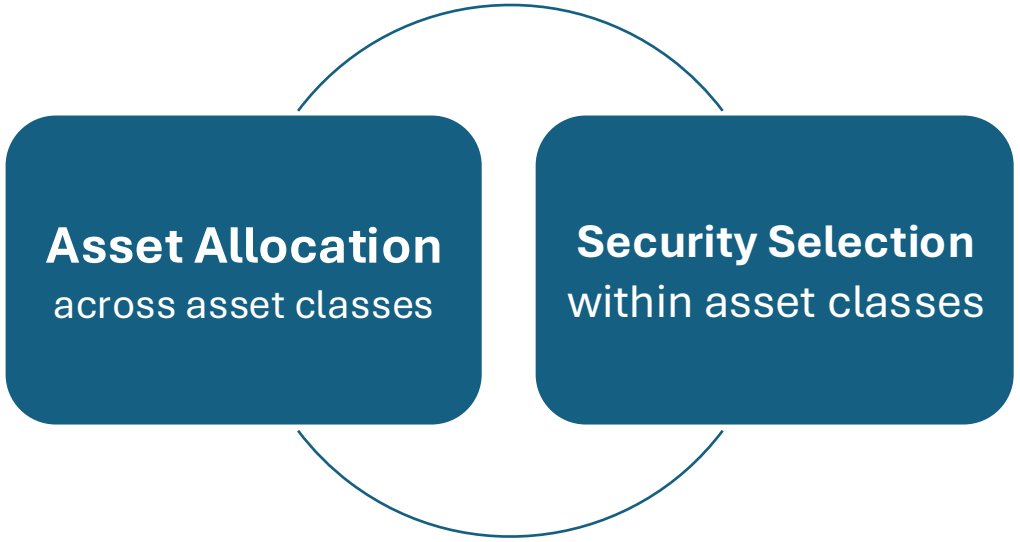
Table: US 10Y Bond and US Equity Comparison

	Total Vol (%)	Carry Vol (%)	Price Vol (%)	Corr(Carry,Price)
US 10Y Bond	7.60	0.37	7.54	0.144
US Equity	15.19	0.19	15.18	0.043

Table: Volatility Decomposition of US 10Y Bond and US Equity

- Carry is stable; price movements drive volatility
- Income and price are weakly correlated
- HY credit not decomposed, “carry” includes default & risk premium, not pure income

Which Matters More?



Conclusion

- Cross-asset differences grow rapidly with horizon
- Long-term performance heavily depends on being in the right asset class
- Allocation effects compound over time

Why Asset Allocation?

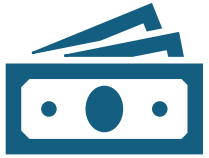
Where Should a CIO Invest Resources?

Invest in:

- **Top-down macro research-** growth, inflation, monetary policy
- **Regime analysis**
- **Dynamic allocation frameworks**

- Special cases in which security selection more important

Single-asset class- credit fund



Restrictions- pensions, endowments

Q7(a): HY vs Equity vs Bonds — Full Risk Profile

	Ann. Geo. Mean (%)	Ann. Vol (%)	Return/Vol	Max Drawdown (%)	Skewness	Excess Kurtosis
US HY Credit	4.92	10.28	0.478	-30.24	0.296	7.602
US Equity	8.39	15.19	0.552	-50.95	-0.489	0.792
US 10Y Bond	3.60	7.60	0.474	-27.29	-0.066	0.891

US HY Credit: Attractive on Paper, Misleading in Practice

- Return: 4.92%
- Volatility: 10.28%

- Max drawdown: -30.24%
- Excess Kurtosis: 7.6

HY = Equity + Credit Risk

Regression	Rolling Betas	HY Behavior
Equity beta ≈ 0.49 **Bond beta ≈ 0.15** **R^2 ≈ 56%**	Equity beta rises in crises Bond beta can turn negative	Risk-on asset (equity-like) Not a diversifier when needed most

Mini-project question

HY Credit as a Risk Premium Strategy

1. Risk Premium Structure

- Earn carry (credit spread) in normal times
- Exposed to large losses in stress (spread widening)
- Payoff = small gains + occasional big losses

2. Time-Varying Attractiveness

- Tight spreads → low compensation → poor risk/reward
- Wide spreads → high compensation → better entry point (but risky)
- → HY requires active timing, not passive holding

3. Constraints & Market Behavior

- Illiquidity in stress → hard to exit positions
- Crowding (carry trade) → sharp unwinds (e.g., 2020)
- Institutional constraints (IG-only) affect who can hold HY